



Anannay Jain
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Communication & Signal Processing

210110021
Dual Degree (B.Tech. + M.Tech.)
Gender: Male
DOB: 13/12/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2026	
Intermediate	CBSE	Blue Heavens Vidyalaya	2021	
Matriculation	CBSE	Maheshwari Public School	2019	

Pursuing a minor degree in **Artificial Intelligence and Data Science** from C-MInDS, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank of 398** in JEE Mains with **100%** ile in **Mathematics** outperforming **1Mn+** applicants ('21)
- Earned Branch Change to Electrical (among **15** freshmen selected) for excellent academic performance ('22)
- Awarded National Talent Search Examination (**NTSE**) scholarship for excelling among **1Mn+** candidates ('19)
- Best Project Award** for SMD reflow station following precise temperature curves in Electronic Design Lab ('24)

PUBLICATIONS

- Anannay Jain**, Shashank Mujumdar et al., "Bridging Language and **Graphs**: Reliable **Text2Cypher** Generation via Schema-Guided Prompting and **Automatic Validation**", submitted to **AAAI 2026** (under review)
- Vashu Chauhan, Shobhit Raj, Shashank Mujumdar, Avirup Saha, **Anannay Jain**, "Mind the Query: A **Benchmark** Dataset towards Text2Cypher Task", submitted to **EMNLP 2025** Industry Track (under review)

PROFESSIONAL EXPERIENCE

IBM Research | AI Research Intern

(May '25 - Aug '25)

Co-authored **two** top-tier conference submissions on advancing Text2Cypher accuracy for graph retrieval

- Boosted Text2Cypher benchmark accuracy by **24%** via a Dynamic Few-Shot + Multi-Stage validation pipeline
- Outperformed **Claude-3.5 Sonnet** (61.58%) by raising **LLaMA-4** accuracy to **64.57%** using dynamic validation
- Elevated CypherBench accuracy from **18.8%** to **77%** by **LoRA-finetuning** Llama-3.1-8B, validating its impact

Aqxe AI | AI/ML Intern

(Dec '24 - Jan '25)

Presented demo in **two** client meetings, securing **Unilever** as the startup's first paying customer

- Delivered faster insights for **multi-hop** QA on tables that have **1701** survey records spread across **nine** sheets
- Designed **interactive** dashboard that translates questions to **graphs** and charts, automating data storytelling
- Achieved **92%** accuracy while reducing code failures by **50%** through a feedback loop in LLM code generation

MapIT.ai | Generative AI Intern

(May '24 - Jul '24)

Received a Letter of Recommendation from the Founder

- Built automated query system using **LlamaIndex**, improving marketing insights and query efficiency by **30%**
- Developed a Q&A system using **RAG** with pandas query engine and GPT-4o, achieving **88%** correct responses
- Used **LangSmith**, reducing deployment time by **15%** through enhanced feedback and automated evaluations

PROJECTS UNDERTAKEN

Real-Time Efficient Object Tracking via Optical Flow | Dual Degree Project

(Aug '24 - Present)

Guide: Prof. Udayan Ganguly, Department of Electrical Engineering

- Boosted tracking throughput **10x** via **YOLOv11m** + stride-based optical flow, with minimal mAP reduction
- Achieved **34.99 FPS** on CPU using **Farneback's** algorithm, ensuring real-time performance without GPUs
- Reduced compute load **9.3x** (**68→7.3 GFLOPs**) via sparse YOLO calls, cutting energy and hardware costs

Sustainable Agriculture to Farm Carbon Credits | Undergraduate Research Program

(Dec '23 - Apr '24)

Guide: Prof. Vikram Vishal, Department of Earth Sciences

- Curated **900+** entries focused on **Carbon Sequestration** strategies in agriculture using **Selenium** and Scrapy
- Combined Regression and Xgboost to predict Carbon Credits with **73%** variance explained & **65%** accuracy
- Provided insights via **Streamlit** UI, increasing potential revenue for farmers by **20%** using predicted strategies

T20 Match Result Predictor | American Express Campus Challenge | Self Project (Jun '24 - Jul '24)

Qualified for Amex Data Challenge Round 2, ranking among IITB's top 15 teams

- Made **60+** features from fundamental variables using **ANOVA**, **Chi-Square** tests to improve model precision
- Utilized **XGBoost** and **SHAP** for feature importance, attaining over **65%** accuracy in predicting outcomes

Metric Learning and Generative AI | Course Project (Oct '23 - Nov '23)

EE782 - Advanced Machine Learning | Guide: Prof. Amit Sethi

- Designed a **Siamese** network using **contrastive loss**, attaining **68%** accuracy for identity differentiation tasks
- Reduced overfitting and stabilized model training using **BatchNorm**, LR scheduling, and gradient clipping
- Generated realistic human faces by training a **DC-GAN**, enriching dataset variety for facial recognition tasks

Predictive Modelling for Vibration Levels | Course Project (Aug '23 - Nov '23)

DS203 - Programming for Data Science | Guide: Prof. Vinay Kulkarni

- Achieved **R² = 0.98** for vibration prediction by developing regression and Random Forest model in sklearn
- Reduced features from **240** to **65** via **p-value** statistical cleaning & iterative elimination, reducing overfitting
- Identified **top-five** controllable parameters using Random Forest feature importance, giving **interpretability**

Limestone Data Challenge | Algorithmic Trading Hackathon (Mar '23 - Apr '23)

Organised by Finance & Analytics Club, IIT Bombay, in association with Tower Research Capital

- Stood among **top 20** teams, earning an exclusive invitation to a closed session with **Tower research** team
- Implemented **Hierarchical Clustering** to classify **100+** stocks into 8 sectors based on variance & return data
- Employed **regression models** to predict and classify **index returns** by using stock data from specific sectors

Deep Learning for Image Classification and Denoising Tasks | Course Project (Feb '24 - Mar '24)

GNR638 - Machine Learning for Remote Sensing - II | Guide: Prof. Biplap Banarjee

- Achieved **68%** accuracy in bird classification across **200** classes using **transfer learning** with **InceptionNetV3**
- Enhanced image clarity via convolutional autoencoder, achieving **24.57 dB PSNR** on blurred test images

Stock Price Prediction using LSTM | Course Project (Oct '23 - Nov '23)

EE782 - Advanced Machine Learning | Guide: Prof. Amit Sethi

- Analyzed min-by-min stock data using **candlestick patterns** by Plotly and applied **EDA** for interday trading
- Trained **LSTM** models on time series stock data using **PyTorch**, improving prediction accuracy with old data
- Created trading module for decisions using **Moving Average**, incorporating bid-ask spread and commissions

POSITIONS OF RESPONSIBILITY

SustLife | Venture Team Lead | Sustainability Platform | IDEAS, IIMB Launchpad (May '24 - Dec '24)

- Ranked in **top 50** of **450+** teams in IDEAS Program, qualifying for **Level 2** receiving **₹0.25Mn** equity-free grant
- Chosen for **NSRCEL IIMB Launchpad's** 27th Cohort, refining venture strategy via 100-day intensive program
- Conducted **50+** interviews with climate change experts for customer discovery & validation to design **MVP**

Pronites | Coordinator | Mood Indigo (Jun '22 - Jan '23)

- Worked in a team of **10**, responsible for smooth **execution** of flagship concerts attended by a crowd of **100K**
- **Conceptualized** and **organized** Livewire - the biggest semi-professional Indie band competition, in **5+** cities

TECHNICAL SKILLS

SKILLS	COMPUTER LANGUAGES	Python, C++, LaTeX, Embedded C, Matlab, Cypher, SQL
	DATA SCIENCE & ML	PyTorch, NumPy, Pandas, Sklearn, Neo4j, Matplotlib
	COMPUTER VISION / NLP	NLTK, OpenCV, LlamaIndex, Huggingface, LangChain
	OS / TOOLS	Git, Docker, Amazon Web Service, Linux (Ubuntu), Bash
ONLINE COURSES	Coursera (Supervised ML), DeepLearning.ai (Agents in Langraph, LangChain, RAG)	

EXTRA-CURRICULAR ACTIVITIES

Sports	• Secured 3rd Position in Revive Sports League (Table Tennis) conducted by Aavhan • Completed one year of military training at the National Cadet Corps (NCC), IIT BOMBAY
Cultural	• Represented Hostel-3 in 15-member dance Squad, securing Second place in hostel GC • Performed Bollywood and Hip-Hop styles at Annual Insync Dance Show, Annual event
Technical	• Completed Corporate Finance Institute courses on Excel & Accounting Fundamentals
Interests	• Enjoy dance, swimming, badminton, and foosball; an admirer of the anime One Piece